MASS MASTER IN ASTROPHYSICS AND SPACE SCIENCE

Forecasting the Exoplanet Yield and Observational Capabilities of the LIFE Mission

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Abtract: The Large Interferometer for Exoplanets (LIFE) mission concept aims to detect and characterize terrestrial exoplanets in the mid-infrared through space-based nulling interferometry. As the mission remains in the simulation and design study phase, forecasting its potential scientific return requires robust modeling of instrument performance and astrophysical populations. This internship project establishes the theoretical and computational background needed to estimate the exoplanet detection yield of LIFE under varying assumptions. The work involves a structured review of mission concept papers; hands-on engagement withexisting LIFE simulation notebooks; and the execution of Monte Carlo frameworks that generate synthetic planetary systems and evaluate their detectability

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Literature Review, Spectral Simulations using LIFEsim & Exoplanet Yield Estimation

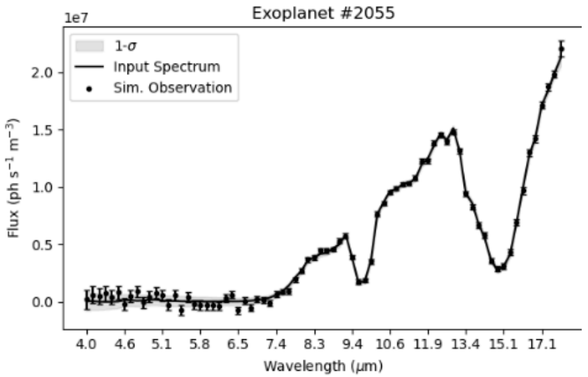
Kammerer & Quanz (2018) — Simulating the exoplanet yield of a space-based mid-infrared interferometer based on Kepler statistics
First LIFE yield simulations
Quanz et al. (2021) — LIFE I: Improved exoplanet detection yield estimates for a large mid-infrared space-interferometer mission
Improved detection yield estimates
Glauser et al. (2024) — The Large Interferometer for Exoplanets (LIFE): a space mission for mid-infrared nulling interferometry
Mission design and instrumentation
Angerhausen et al. (2023) — LIFE VIII: Where is the phosphine? Observing exoplanetary PH_3
Atmospheric biosignature detection
Alei et al. (2024) — LIFE XIII: The value of combining thermal emission and reflected light for the characterization of Earth twins
Thermal + reflected characterization
Angerhausen et al. (2025) — What if we find nothing? Bayesian analysis of the statistical information of null results...
Bayesian interpretation of null detections
Kammerer et al. (2022) — LIFE VI: Detecting rocky exoplanets in the habitable zones of Sun-like stars
Habitable rocky planet detection

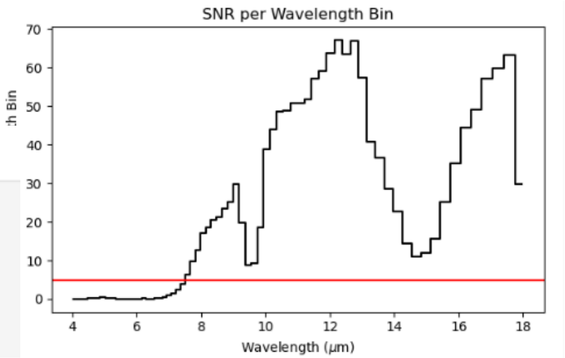
LIFE's science return depends on: occurrence rates, dust levels, instrument performance, observing strategy, and statistical interpretation.

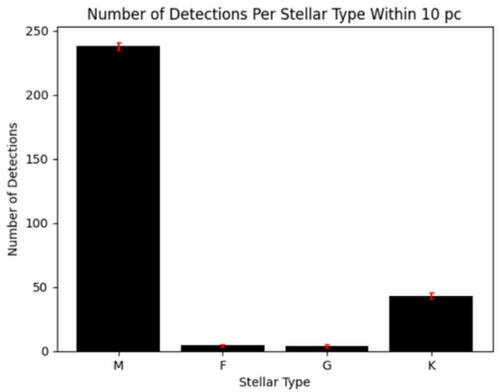
- Simulated 540 mid-infrared spectra (4–18.5 μm)
- Each planet → synthetic flux + SNR spectrum

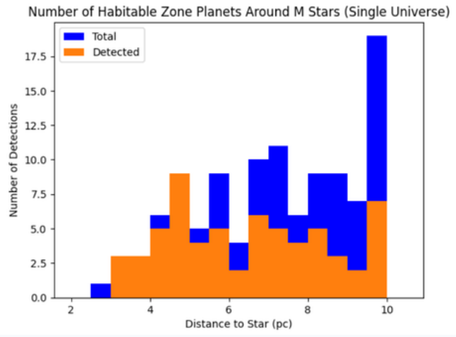
Simulation variables

```
scenario = 'baseline' # Options: 'basline', 'pessimistic', 'optimistic'
spectral_resolution = 50  # Resolving power
minimum_wavelength = 4 # Minimum wavelength of the spectrometer in micron
maximum_wavelength = 18 # Maximum wavelength of the spectrometer in micron
integration_time = 55*60*60  # Overall integration time in s
exozodi_level = 3  # Exozodi level in zodis
```







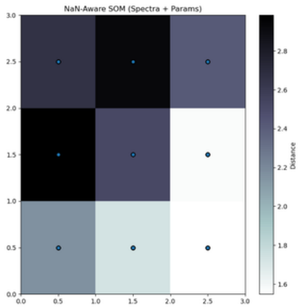


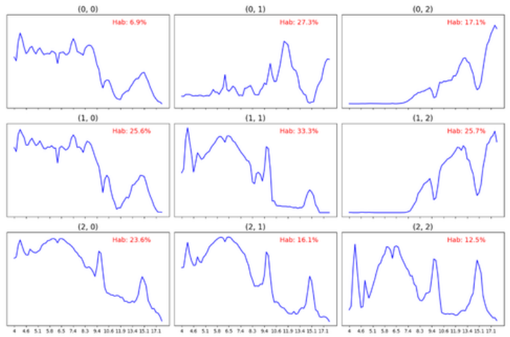
Mean detected planets computed per stellar type
- M-type stars dominate yield
- Smaller habitable zone separations

Results consistent with:
- Kammerer & Quanz (2018)
- Quanz et al. (2021)

The analysis confirms that a significant fraction of detectable terrestrial planets are expected around low-mass stars.

SOMs -Self-Organizing Maps

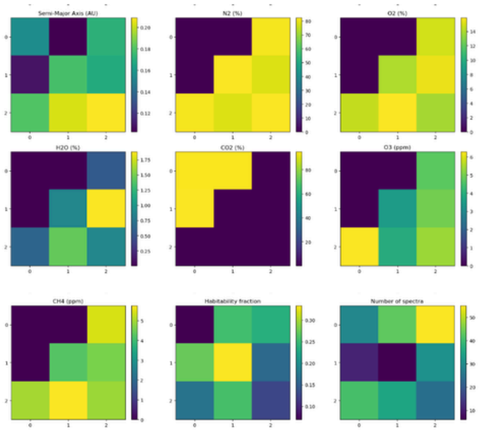


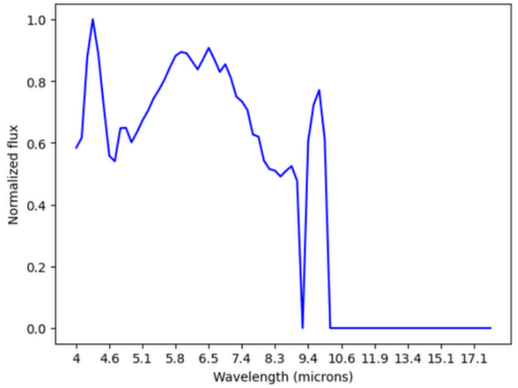


Left: NaN-aware SOM trained using spectral and physical parameters.
Right: Average spectra associated with each SOM cell.

Conclusions

- **Yield** strongly depends on:
  - Instrument performance
  - Astrophysical assumptions
- **M-type stars** dominate detections within 10 pc
- **G-type HZ planets** remain scientifically important
- **SOMs** are powerful for **interpreting spectra**
- **LIFE** shows strong potential for:
  - Detecting terrestrial planets
  - **Characterizing atmospheric properties**





In habitable zone!

Parameters planet:

Temperature: 228 K

Mass: 5.15 $M_\oplus$

Radius: 1.49 $R_\oplus$

Parameters host star:

Temperature: 3100 K

Mass: 0.18 $M_\odot$

Radius: 0.24 $R_\odot$

Semi-major axis: 0.1 AU

Future thesis work

Forecast the **exoplanet yield of PLATO Mission** and compare the performance with LIFE Mission !!!

Thank You!!!